KAIZEN MATHS

# Lesson starter

**Work, Energy and Power**

Modelling Problems

Name

**S1**

A particle of mass 5 kg slides from rest down a smooth slope of length 8 m inclined at 30° to the horizontal. Find its speed at the bottom. Take g = 9.8 m $s^{-2}$.

**S2**

A particle of mass 10 kg slides from rest down a rough slope of length 6 m inclined at 25°. The friction force is 8 N. Find its speed at the bottom. Take g = 9.8 m $s^{-2}$.

**S3**

A ball is projected vertically upwards with speed 24 m $s^{-1}$. Find the greatest height reached above the point of projection. Take g = 9.8 m $s^{-2}$.

**S4**

A vehicle of mass 800 kg moves up a slope inclined at 12° at a constant speed of 10 m $s^{-1}$. The resistance to motion is 100 N. Find the power of the engine. Take g = 9.8 m $s^{-2}$.



Show your working

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MEC-8E86-L3 · S

Show your working

KAIZEN MATHS

# Exit ticket

**Work, Energy and Power**

Modelling Problems

Name _____

**EA1**

A vehicle of mass 100 kg accelerates from 4 m $s^{-1}$ to 10 m $s^{-1}$ over 120 m on a horizontal road. The resistance is 20 N. Find the constant driving force.

**EA2**

A particle of mass 2 kg slides from rest down a smooth slope of length 10 m inclined at 40° to the horizontal. Find its speed at the bottom. Take g = 9.8 m $s^{-2}$.



Show your working

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# Exit ticket

**Work, Energy and Power**

Modelling Problems

Name _____

**EB1**

A particle of mass 8 kg slides from rest down a rough slope of length 10 m inclined at 40°. The friction force is 6 N. Find its speed at the bottom. Take g = 9.8 m $s^{-2}$.



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# Exit ticket

**Work, Energy and Power**

Modelling Problems

Name _____

**EA1**

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**EA2**

A particle of mass 2 kg slides from rest down a smooth slope of length 10 m inclined at 40° to the horizontal. Find its speed at the bottom. Take g = 9.8 m $s^{-2}$.

MEC-8E86-L3 · EA

Show your working

KAIZEN MATHS

# Exit ticket

**Work, Energy and Power**

Modelling Problems

Name _____

**EB1**

A particle of mass 8 kg slides from rest down a rough slope of length 10 m inclined at 40°. The friction force is 6 N. Find its speed at the bottom. Take g = 9.8 m $s^{-2}$.

MEC-8E86-L3 · EB

Show your working

# Teacher answers & working

**Work, Energy and Power · Modelling Problems**

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## S1

A particle of mass 5 kg slides from rest down a smooth slope of length 8 m inclined at 30° to the horizontal. Find its speed at the bottom. Take g = 9.8 m $s^{-2}$.

**Answer:**

$8.854\ \text{m s}^{-1}$

**Working:**

On a smooth slope, loss of gravitational potential energy becomes gain in kinetic energy.

Vertical drop h = 8 sin 30° = 4 m

$mgh = \frac{1}{2}mv^2$

$gh = \frac{1}{2}v^2$

$v^2 = 2gh$

$v^2 = 2 \times 9.8 \times 4 = 78.4$

$v = \sqrt{(78.4)} = 8.854\ \text{m s}^{-1}$

## S2

A particle of mass 10 kg slides from rest down a rough slope of length 6 m inclined at 25°. The friction force is 8 N. Find its speed at the bottom. Take g = 9.8 m $s^{-2}$.

**Answer:**

$6.332\ \text{m s}^{-1}$

**Working:**

Gain in kinetic energy equals work done by the component of weight minus work done against friction.

Component of weight down the slope = mg sin 25°

$mg \sin 25° = 10 \times 9.8 \times \sin 25° = 41.417\ \text{N}$

Net work done = (41.417 − 8) × 6 = 200.5 J

$\frac{1}{2}mv^2 = 200.5$

$\frac{1}{2} \times 10 \times v^2 = 200.5$

$v^2 = \frac{400.999}{10} = 40.1$

$v = 6.332\ \text{m s}^{-1}$

# Teacher answers & working

**Work, Energy and Power · Modelling Problems**

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## S3

A ball is projected vertically upwards with speed 24 m $s^{-1}$. Find the greatest height reached above the point of projection. Take g = 9.8 m $s^{-2}$.

**Answer:**

29.388 m

**Working:**

At the greatest height, all the initial kinetic energy has become gravitational potential energy.

$\frac{1}{2}mu^2 = mgh$

$\frac{1}{2}u^2 = gh$

$h = \frac{u^2}{2g}$

$h = \frac{24^2}{2 \times 9.8}$

$h = 29.388$ m

## S4

A vehicle of mass 800 kg moves up a slope inclined at 12° at a constant speed of 10 m $s^{-1}$. The resistance to motion is 100 N. Find the power of the engine. Take g = 9.8 m $s^{-2}$.

**Answer:**

17300.277 W

**Working:**

At constant speed, the engine balances the component of weight down the slope and the resistance.

Component of weight down the slope = mg sin 12°

$mg \sin 12° = 800 \times 9.8 \times \sin 12° = 1630.028$ N

Driving force = 1630.028 + 100 = 1730.028 N

$P = Fv$

$P = 1730.028 \times 10$

$P = 17300.277$ W

# Teacher answers & working

**Work, Energy and Power · Modelling Problems**

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## EA1

A vehicle of mass 100 kg accelerates from 4 m $s^{-1}$ to 10 m $s^{-1}$ over 120 m on a horizontal road. The resistance is 20 N. Find the constant driving force.

**Answer:**

55 N

**Working:**

Driving work minus work done against resistance equals gain in kinetic energy.

$Fs - Rs = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

$(F - 20) \times 120 = \frac{1}{2} \times 100 \times (10^2 - 4^2)$

Gain in kinetic energy = 4200 J

$(F - 20) \times 120 = 4200$

$F - 20 = \frac{4200}{120} = 35$

$F = 55$ N

## EA2

A particle of mass 2 kg slides from rest down a smooth slope of length 10 m inclined at 40° to the horizontal. Find its speed at the bottom. Take g = 9.8 m $s^{-2}$.

**Answer:**

11.224 m $s^{-1}$

**Working:**

On a smooth slope, loss of gravitational potential energy becomes gain in kinetic energy.

Vertical drop h = 10 sin 40° = 6.428 m

$mgh = \frac{1}{2}mv^2$

$gh = \frac{1}{2}v^2$

$v^2 = 2gh$

$v^2 = 2 \times 9.8 \times 6.428 = 125.986$

$v = \sqrt{(125.986)} = 11.224$ m $s^{-1}$

# Teacher answers & working

**Work, Energy and Power · Modelling Problems**

MEC-8E86-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

**EB1**

A particle of mass 8 kg slides from rest down a rough slope of length 10 m inclined at 40°. The friction force is 6 N. Find its speed at the bottom. Take g = 9.8 m $s^{-2}$.

**Answer:**

10.535 m $s^{-1}$

**Working:**

Gain in kinetic energy equals work done by the component of weight minus work done against friction.

Component of weight down the slope = mg sin 40°

$mg \sin 40° = 8 \times 9.8 \times \sin 40° = 50.395$ N

Net work done = (50.395 − 6) × 10 = 443.945 J

$\frac{1}{2}mv^2 = 443.945$

$\frac{1}{2} \times 8 \times v^2 = 443.945$

$v^2 = \frac{887.891}{8} = 110.986$

$v = 10.535$ m $s^{-1}$